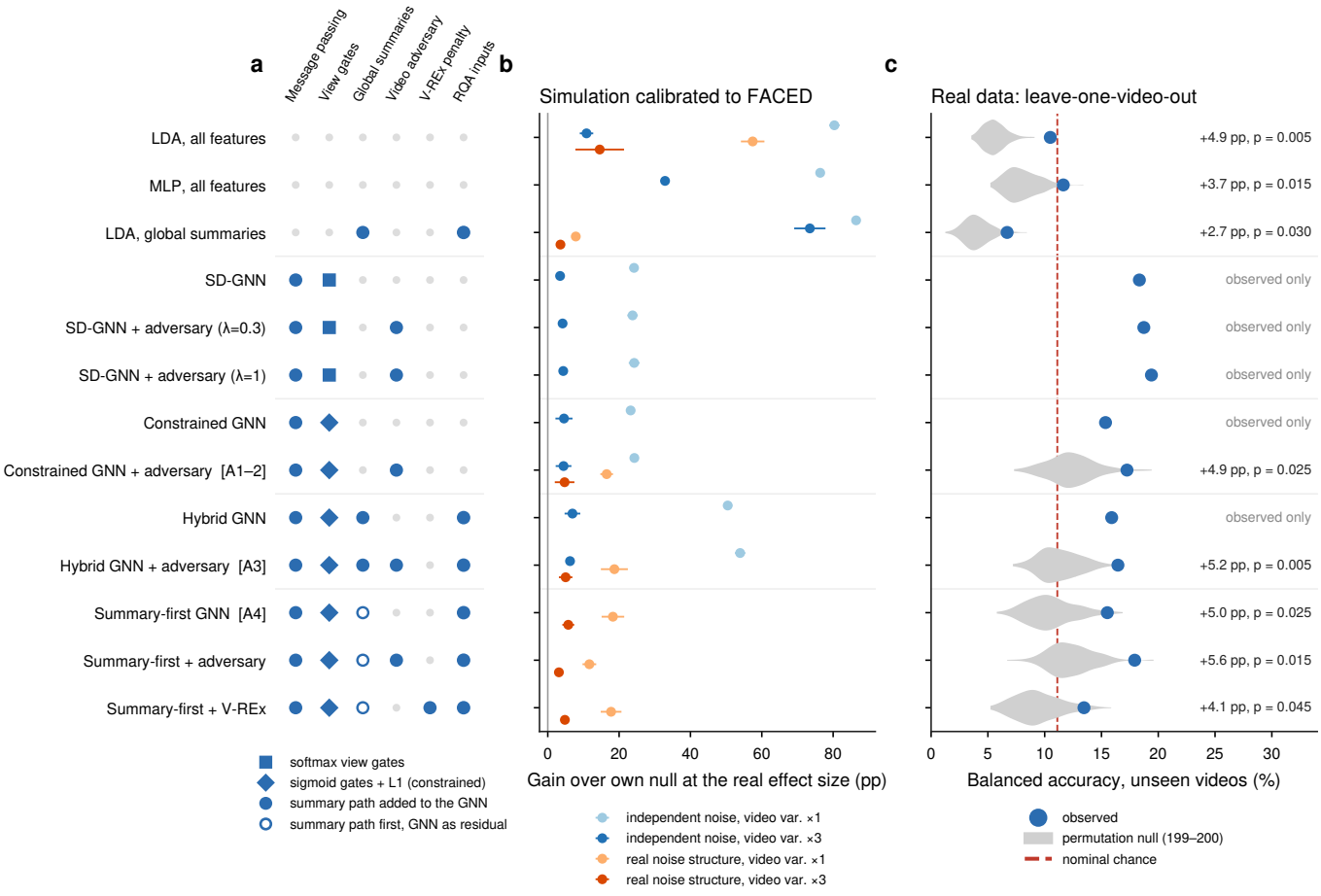
a
Message passing
View gates
Global summaries
Video adversary
V-REx penalty
RQA inputs
LDA, all features
MLP, all features
LDA, global summaries
SD-GNN
SD-GNN + adversary (λ=0.3)
SD-GNN + adversary (λ=1)
Constrained GNN
Constrained GNN + adversary [A1–2]
Hybrid GNN
Hybrid GNN + adversary [A3]
Summary-first GNN [A4]
Summary-first + adversary
Summary-first + V-REx
softmax view gates
sigmoid gates + L1 (constrained)
summary path added to the GNN
summary path first, GNN as residual
b
Simulation calibrated to FACED
0
20
40
60
80
Gain over own null at the real effect size (pp)
independent noise, video var. ×1
independent noise, video var. ×3
real noise structure, video var. ×1
real noise structure, video var. ×3
c
Real data: leave-one-video-out
+4.9 pp, p = 0.005
+3.7 pp, p = 0.015
+2.7 pp, p = 0.030
observed only
observed only
observed only
observed only
+4.9 pp, p = 0.025
observed only
+5.2 pp, p = 0.005
+5.0 pp, p = 0.025
+5.6 pp, p = 0.015
+4.1 pp, p = 0.045
0
5
10
15
20
25
30
Balanced accuracy, unseen videos (%)
observed
permutation null (199–200)
nominal chance